

LearnPulse AI LMS — Week 11 Report

Supplemental AI Learning Features Implementation & Final Verification Report

Executive Summary

This report documents the completed implementation and final verification of **Week 11: Supplemental AI Learning Features** for the LearnPulse AI-Powered LMS backend. Week 11 introduces four supplemental learning endpoints built upon Spring AI (0.8.1), Ollama (qwen2.5-coder:7b), PostgreSQL (postgres-pgvector), and JWT Security.

Verification was confirmed through a **149-test automated regression suite (149 passed, 0 failures, 0 errors, 0 skipped, BUILD SUCCESS)** and a **live end-to-end (E2E) verification execution** against the running Spring Boot backend and live local Ollama/Qwen server.

1. Scope & Feature Specifications

- **Document Summarization (POST /api/ai/summarize)**

Analyzes raw educational text up to 20,000 chars. Produces structured summary, key takeaways, and keywords. Handled by DocumentSummarizationService.

- **Source Code Explainer (POST /api/ai/explain-code)**

Performs static analysis on code snippets up to 10,000 chars. Provides purpose, logic, variables, methods, flow, algorithm, and $O(1)$ complexity. Submitted code is treated strictly as **DATA ONLY** inside [SOURCE CODE BOUNDARY START] fences — **NEVER compiled or executed**. Handled by CodeExplanationService.

- **AI Study Planner (POST /api/ai/study-plan)**

Generates daily schedules for exam prep (max 10 subjects). Backend deterministically calculates days remaining. Post-LLM rule: **allocated daily hours** $\leq$ **availableHoursPerDay**. Violations are rejected cleanly via ApiException (HTTP 500) rather than silently clamped. Handled by StudyPlanService.

- **AI Flashcards (POST /api/ai/flashcards)**

Generates Q&A; flashcards (\$1 \leq \text{count} \leq 20\$) from material up to 15,000 chars. Performs post-LLM deterministic backend question deduplication before returning response. Handled by FlashcardGenerationService.

2. Architecture & Design Decisions

System Flow: Student $\rightarrow$ Spring Boot AI Endpoints (AiFeaturesController) $\rightarrow$ Feature Services $\rightarrow$ PromptManager / ChatClient $\rightarrow$ Ollama (qwen2.5-coder:7b) $\rightarrow$ StructuredAiOutputParser $\rightarrow$ Validated DTO Response.

StructuredAiOutputParser Capabilities: Strips markdown code fences (``json ... ``), extracts JSON from prose, deserializes using Jackson with JavaTimeModule for LocalDate ISO parsing, evaluates Jakarta Bean Validations (@Valid), and produces standardized error handling without exposing raw AI output.

3. Automated Test Suite Results (mvn test)

The complete LearnPulse regression test suite (Weeks 1 through 11) was executed via Maven with OpenJDK 21:

Metric	Result
Total Tests Executed	149 (increased from 143; 6 added auth & LLM-failure tests)
Passed	149
Failures	0
Errors	0
Skipped	0
Maven Status	BUILD SUCCESS
Execution Time	53.637 seconds

LLM Failure & Security Test Coverage:

- **LLM Failure Handling:** LLM failure handling is explicitly tested across all four Week 11 AI services, with standardized 503 Service Unavailable handling.
- **Unauthenticated Requests:** Missing JWT token returns HTTP **401 Unauthorized**.
- **Non-STUDENT Access:** Authenticated requests from non-STUDENT roles (TEACHER) return HTTP **403 Forbidden** across all Week 11 endpoints.
- **STUDENT Access:** Authenticated STUDENT requests proceed successfully.

4. Live End-to-End (E2E) Verification Results

Executed via verify_week11_ai.py against running Spring Boot backend, postgres-pgvector database (learning_assistant_db), and live local Ollama (qwen2.5-coder:7b):

Test Case / Endpoint	Live Result	Latency	Details Verified
Security: 401 Unauthenticated	PASS	4.5 ms	Missing token rejected with 401
Security: 403 Non-STUDENT	PASS	14.6 ms	TEACHER role rejected with 403
Validation: 400 Blank Content	PASS	11.0 ms	Blank input rejected with 400
POST /api/ai/summarize	PASS	47,621.5 ms	322-char summary, 5 takeaways, 4 keywords
POST /api/ai/explain-code	PASS	73,853.1 ms	Purpose, step-by-step logic, O(1) complexity
POST /api/ai/study-plan	PASS	133,449.9 ms	7 daily schedules, hours limit validated
POST /api/ai/flashcards	PASS	18,788.0 ms	2 Q&A; cards, deduplication verified

5. Compliance & Project Boundaries

- **Project Boundary:** Verification used exclusively the LearnPulse project codebase (/Users/krishnagarg/lms internship).
- **Infrastructure Isolation:** Connected exclusively to LearnPulse PostgreSQL (postgres-pgvector / learning_assistant_db) and local Ollama instance.
- **Zero SmartOnboard Contact:** No SmartOnboard resources were modified, started, stopped, accessed, or used.
- **Git Hygiene:** No Git commits or pushes were performed.

Week 11 is fully implemented and verified through the complete 149-test regression suite and live end-to-end verification of all four AI endpoints.